

Spatially Autoregressive Models

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Glossary

Auto-Model A phrase coined by Besag that describes a probability model where attribute values for a single variable distributed across n locations covary with weighted sums of nearby, neighboring attribute values for the same variable, and whose conditional probabilities can be written in terms of these weighted sums.

Autoregression A regression model specification that includes memory or feedback, in terms of weighted sums of attribute values, such that the system of observations that it describes is interdependent and contains important internal structure, and the response variable appears on both sides of the equation.

Generalized Linear Model A model specification in which some nonlinear transformation of a response variable is expressed as the weighted sum of a set of covariates whose weights are coefficients computed from observed data, without including an error term (the random component is a function of the mean response and is generated by the underlying probability distribution), and whose variation may be assumed to be non-normally distributed (e.g., Poisson, binomial, negative binomial) according to the mathematical link between the transformation and the non-normality.

Georeferenced An ordering and then tagging to a two-dimensional surface (e.g., the Earth's surface) of observations with Cartesian coordinates, such as longitude and latitude or Universal Transverse Mercator (UTM) coordinates, as symbolic descriptors of place.

Linear Model A model specification in which a response variable is expressed as the weighted sum of a set of covariates whose weights are coefficients computed from observed data, to which a random error term is added whose variation is assumed to be normally distributed, where scatterplots for response variables and each covariate approximate a straight-line graph.

Maximum Likelihood A popular inferential statistical technique that yields parameter estimates for a selected probability distribution by maximizing the joint probability for n observed values in a given dataset – the product of n individual probabilities for independent observations – resulting in estimates that make the observed data most likely.

MCMC (Markov chain Monte Carlo) An iterative mechanism for drawing dependent samples in situations where regular sampling is difficult, if not completely impossible, in order to compute simultaneous likelihood results, that generates Markov chains with Monte Carlo

simulation, exploits sufficient statistics, makes use of marginal probabilities, and frequently utilizes pseudo-likelihood results.

Normalizing Constant A constant by which all possible probabilities of a probability model must be multiplied to ensure that they sum to 1, for discrete values, and integrate to 1, for continuous values.

Spatial Autocorrelation The self-correlation of values of a given attribute arising from their nearby locations in geographic space, and the essence of the first law of geography.

Spatial Filter A model specification that decomposes a georeferenced response variable into the sum of two synthetic variates, one containing spatial autocorrelation and one free of spatial autocorrelation. These former synthetic variates can be a linear combination of the eigenvectors of a geographic weights matrix.

Background: The Literal Meaning of Autoregression

The prefix 'auto' in the term 'autoregressive' (AR) means 'self' and the root 'regressive' means regression. In other words, a variable containing n observed attribute values is being regressed on itself, which results in the response variable, Y , whose values often are denoted by the n -by-1 vector $\mathbf{Y}$, where n is the number of observations, appearing on both sides of the equal sign. The selection of values of Y to be on the right-hand side of an equation is by observation, which is locational in the spatial context, and is dictated by some rule in such a way that each value of Y is not being regressed on itself, but rather on some subset of the remaining $n - 1$ values of Y . For geographic models, the adjective 'spatial' dictates that these subsets comprise neighboring values of each location. Because of their closeness in geographic space, neighboring values tend to contain duplicated information. This situation is similar to taking a sample and falsely doubling its size merely by duplicating each of its entries. Spatially redundant information needs to be accounted for by regressing Y on neighboring values of Y . Otherwise, results can be misleading, as would be the case for the artificially doubled sample. Differentiation of neighboring and non-neighboring values frequently is achieved with n binary 0 – 1 indicator variables (0 if the locations of Y values are not neighboring, and 1 if the locations are), each containing n entries. These n indicator variables then are

collected into an n -by- n table, commonly called a geographic weights matrix, and usually denoted by matrix $\mathbf{C}$. Consequently, in matrix form, vector $\mathbf{Y}$ is on the left-hand side and vector $\mathbf{CY}$ is on the right-hand side of an AR equation. Because vector $\mathbf{CY}$ is the sum of surrounding neighboring observed values of variable Y , it frequently is replaced with its row-standard version (i.e., each cell entry in matrix $\mathbf{C}$ is divided by its corresponding row sum – the number of 1s in its row), denoted by $\mathbf{WY}$, which is the average of the surrounding values of Y . Individual i th observation row entries for these two versions are denoted, respectively, by $\mathbf{C}_i\mathbf{Y}$ and $\mathbf{W}_i\mathbf{Y}$.

Introduction: The Meaning of Spatial Correlation

Correlation refers to the relationship between two variables, X and Y , can be quantified with a Pearson product-moment coefficient, and often is portrayed with a scatter diagram (a two-dimensional plotting of individual observation value pairs, (X_i, Y_i) , where i denotes an observation's name). Positive linear correlation indexes a tendency for high values of X to be paired with high values of Y , medium values of X with medium values of Y , and low values of X with low values of Y . As this tendency becomes more pronounced, these pairings increasingly align along a line – a straight line when a relationship is linear. Negative linear correlation indexes a tendency for high values of X to be paired with low values of Y , medium values of X with medium values of Y , and low values of X with high values of Y . Finally, no correlation (i.e., randomness) indexes a tendency for high values of X to be paired with high, medium, and low values of Y , medium values of X with high, medium, and low values of Y , and low values of X with high, medium, and low values of Y . Conventional linear regression models capture this correlation.

Spatial correlation – also known as spatial autocorrelation – replaces each value of X with $\mathbf{C}_i\mathbf{Y}$ or $\mathbf{W}_i\mathbf{Y}$ in this preceding relationship scenario, can be quantified with a Moran coefficient or a Geary ratio, and often is portrayed with a Moran scatterplot or a semivariogram plot, which are customized scatterdiagrams. Accordingly, positive spatial correlation indexes a tendency for high values of, say, $\mathbf{W}_i\mathbf{Y}$ to be paired with high values of Y , medium values of $\mathbf{W}_i\mathbf{Y}$ with medium values of Y , and low values of $\mathbf{W}_i\mathbf{Y}$ with low values of Y . Negative spatial correlation indexes a tendency for high values of $\mathbf{W}_i\mathbf{Y}$ to be paired with low values of Y , medium values of $\mathbf{W}_i\mathbf{Y}$ with medium values of Y , and low values of $\mathbf{W}_i\mathbf{Y}$ with high values of Y . Finally, no spatial correlation indexes a tendency for high values of $\mathbf{W}_i\mathbf{Y}$ to be paired with high, medium, and low values of Y ; medium values of $\mathbf{W}_i\mathbf{Y}$ with high, medium, and low values of Y ; and low values

of $\mathbf{W}_i\mathbf{Y}$ with high, medium, and low values of Y . Most spatial correlation is moderate and positive; one exception is spatial correlation for remotely sensed images, which almost always is very strong and positive. Spatial AR models capture this spatial correlation.

The Auto-Normal Probability Model

Some interval/ratio georeferenced datasets have frequency distributions that mimic a bell-shaped curve (e.g., geo-coding errors). One principal, and perhaps the most conspicuous, impact of spatial autocorrelation in terms of this type (as well as other types) of variable is on variation. The overwhelming majority case of positive spatial autocorrelation results in variance inflation, a well-known feature of georeferenced data, which for a normal frequency distribution means the central frequencies (i.e., numbers of values near the mean) decrease and the tail frequencies increase – peakedness (i.e., kurtosis) is altered because the curve is flattened. Less well known is that the intermediate frequencies can become disproportionately reduced in order for increased concentration very close to the mean (see Figure 1). Spatial auto-normal AR models seek to furnish a description of these data.

Three spatial linear regression model specifications have become favored in the literature to describe georeferenced data, namely the

conditional AR (CAR) model:

$$\mathbf{Y} = (\mathbf{I} - \mathbf{D})\mathbf{Y} + \mathbf{DX}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \mathbf{D}^T\mathbf{D} = (\mathbf{I} - \rho\mathbf{C})\mathbf{M} \quad [1]$$

AR response model:

$$\mathbf{Y} = \rho\mathbf{WY} + \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \quad [2]$$

and simultaneous AR (SAR) model:

$$\mathbf{Y} = \rho\mathbf{WY} + (\mathbf{I} - \rho\mathbf{W})\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \quad [3]$$

where $\mathbf{X}$ is an n -by- $(p+1)$ matrix of covariates, $\boldsymbol{\beta}$ is a $(p+1)$ -by-1 vector of regression coefficients, ρ is the spatial AR parameter, $\mathbf{M}$ is an n -by- n diagonal matrix, often set equal to the n -by- n identity matrix $\mathbf{I}$, and $\boldsymbol{\varepsilon}$ is an n -by-1 vector of independent and identically normal distribution random errors. The first of these specifications describes spatial autocorrelation in terms of a first-order inverse covariance structure (i.e., $(\mathbf{I} - \rho\mathbf{C})$, which involves the first-order term $\mathbf{C}$); the other two describe it in terms of a second-order structure (i.e., $(\mathbf{I} - \rho\mathbf{W})'(\mathbf{I} - \rho\mathbf{W})$, where prime denotes the matrix transpose operation, which involves the second-order term $\mathbf{W}'\mathbf{W}$). A first-order model represents weaker spatial autocorrelation over neighboring locations within a relatively short-distance band; a second-order model

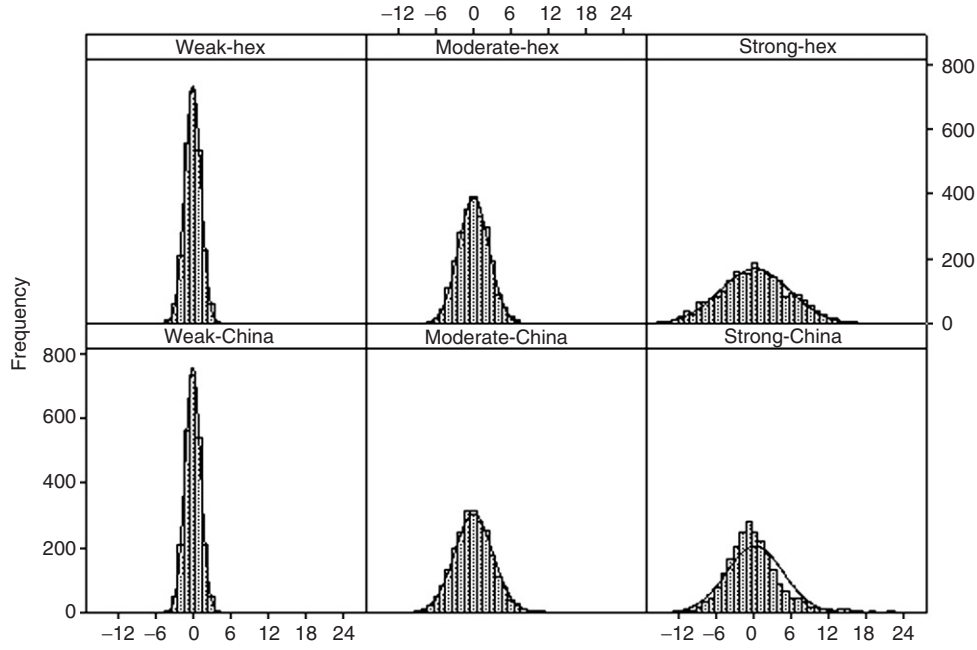


Figure 1 Histograms for simulated normal data containing varying levels of positive spatial autocorrelation: $\mu=5$ and $\sigma=1$. Top: a 45-by-54 regular hexagonal tessellation ($n=2430$) and bottom: China's 1990 county irregular surface partitioning ($n=2379$).

represents stronger spatial autocorrelation over neighboring locations in a greater-distance band. The AR model, called the lag model in spatial econometrics, is a special case of the SAR model in which the spatial lag component coefficients for the covariates are restricted to zero rather than set to ρ . The AR specification conceptualizes spatial autocorrelation as a direct effect outcome; for example, high population density in a given location tends to promote high population density in nearby locations. In contrast, the CAR and SAR specifications conceptualize spatial autocorrelation as an indirect, error effect; for example, a rain storm that moved across a field of crops may well create spatial patterning that is not captured by a crop yield model specification. Spatial scientist practitioner preferences tend to reflect employment of the CAR model in remote sensing, the AR model in spatial econometrics, and the SAR model in spatial statistics. One deviation from this usage pattern is employment of the CAR model in hierarchical Bayesian modeling by spatial statisticians, and hence epidemiologists who engage in disease mapping.

Parameter estimation most often is achieved using maximum likelihood techniques, which optimize the observed data likelihood function. If the spatial covariation is denoted by matrix $\mathbf{V}$ – that is, $\mathbf{V}^{-1} = (\mathbf{I} - \rho\mathbf{C})$ for the CAR model with $\mathbf{M} = \mathbf{I}$, and $\mathbf{V}^{-1} = (\mathbf{I} - \rho\mathbf{W})'(\mathbf{I} - \rho\mathbf{W})$ for the SAR model – then maximum likelihood estimates are given by

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1}\mathbf{Y} \quad [4]$$

$$\hat{\sigma}^2 = (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}})' \mathbf{V}^{-1}(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}})/n \quad [5]$$

and $\hat{\rho}$ is yielded by MIN:

$$\begin{aligned} & [(e^{-J(\rho)})^2 (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}})' \mathbf{V}^{-1}(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}})]^{-n/2} \\ & \text{over the range } \left(\frac{1}{\lambda_{\min}}, \frac{1}{\lambda_{\max}} \right) \end{aligned} \quad [6]$$

where $\mathbf{V}^{-1}$ is a function of ρ , and $J(\rho)$ refers to the calculus concept ‘Jacobian of a transformation’, which ensures that all probabilities integrate to 1. Meanwhile, for the AR model,

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1/2}\mathbf{Y} \quad [7]$$

$$\hat{\sigma}^2 = \mathbf{Y}'(\mathbf{V}^{-1/2})'[\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']\mathbf{V}^{-1/2}\mathbf{Y}/n \quad [8]$$

and $\hat{\rho}$ is yielded by MIN:

$$[(e^{-J(\rho)})^2 (\mathbf{V}^{-1/2}\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}})'(\mathbf{V}^{-1/2}\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}})]^{-n/2} \quad [9]$$

where $\mathbf{V}^{-1/2}$ commonly is $(\mathbf{I} - \rho\mathbf{W})$.

Equation [4] renders unbiased estimates whose expectations are equivalent to its ordinary least-squares (OLS) counterparts (i.e., $\mathbf{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$); OLS first moments often are unbiased in the presence of spatial autocorrelation. Equation [7] renders biased OLS estimates (i.e., $\mathbf{E}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1/2}\mathbf{X}\boldsymbol{\beta}$). The standard bias correction adjustments for eqns [5] and [8] are to divide by $(n-p-1)$, for p covariates, rather than just n ; but this adjustment does not render unbiased OLS estimates in

the presence of spatial autocorrelation without also adjusting for the autocorrelation – this is where spatial autocorrelation introduces variance inflation, and where statistical efficiency is impacted upon. In ideal situations these estimates contain sufficient statistics, with the set of complete sufficient statistics now including a spatial covariation term.

An alternative specification to eqns [1]–[3] exploits the geographic structure latent in matrix $\mathbf{C}$. Pre- and post-multiplying this geographic weights matrix by the multivariate statistics projection matrix $(\mathbf{I} - \mathbf{1}\mathbf{1}'/n)$ coincides with a centering of the data and yields the Moran coefficient numerator term $(\mathbf{I} - \mathbf{1}\mathbf{1}'/n)\mathbf{C}(\mathbf{I} - \mathbf{1}\mathbf{1}'/n)$. Now the right-hand side of a regression equation can be specified using the matrix decomposition $(\mathbf{I} - \mathbf{1}\mathbf{1}'/n)\mathbf{C}(\mathbf{I} - \mathbf{1}\mathbf{1}'/n) = \mathbf{E}\mathbf{\Lambda}\mathbf{E}'$, where $\mathbf{E}$ and $\mathbf{\Lambda}$, respectively, are the matrix of eigenvectors and the diagonal matrix of eigenvalues of this modified matrix $\mathbf{C}$. This projection matrix essentially replaces the principal eigenfunction with one whose eigenvalue is 0 and whose eigenvector is 1. Judiciously selecting from among the n eigenvectors yields a subset of them, say $\mathbf{E}_K$, which can be used to replace AR terms like $\rho\mathbf{C}$ and $\rho\mathbf{W}$ in eqns [1]–[3]. Accordingly, the spatial filter (SF) model may be written as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta}_X + \mathbf{E}_K\boldsymbol{\beta}_E + \boldsymbol{\varepsilon} \quad [10a]$$

where the subset of eigenvectors $\mathbf{E}_K$, which are mutually orthogonal and uncorrelated, can be selected with a stepwise linear regression procedure, $\boldsymbol{\beta}_X$ and $\boldsymbol{\beta}_E$, respectively, are regression coefficients for the p covariates and the K selected eigenvectors, and spatial autocorrelation detected in non-SF OLS residuals is captured by $\mathbf{E}_K\boldsymbol{\beta}_E$, rendering the resulting residual vector, $\mathbf{e}$, void of spatial autocorrelation. An appealing feature of this specification is that distribution theory for the Moran coefficient is known for the resulting linear regression residual vector $\mathbf{e}$ (e.g., a sampling distribution mean of

$$\frac{n}{\mathbf{1}^T\mathbf{C}\mathbf{1}} \frac{\text{TR}\{[(\mathbf{X}\mathbf{E}_K)'(\mathbf{X}\mathbf{E}_K)]^{-1}(\mathbf{X}\mathbf{E}_K)'\mathbf{C}(\mathbf{X}\mathbf{E}_K)\}}{n-1-p-K} \quad [10b]$$

where TR denotes the matrix trace operation, and an approximate standard deviation of $\sqrt{2/\mathbf{1}^T\mathbf{C}\mathbf{1}}$.

A simple illustration of how this spatial statistical modeling may alter the understanding of empirical situations can be gleaned by revisiting Haining's 1991 Glasgow health data. The bivariate correlation between respiratory and cancer death rates across the 87 health districts is 0.63. Both variables contain weak-to-moderate positive spatial autocorrelation, respectively, exhibiting roughly 1/4 and 1/3 redundant information. Results from SAR models indicate that this correlation actually is 0.51, a significantly lower value. Results from SF models indicate a value of 0.45, and reveal that spatial

autocorrelation in the two variables has two map patterns in common, which inflates the correlation beyond 0.63, coupled with two and five map patterns that are unique to, respectively, respiratory and cancer death rates, which deflate the correlation to 0.63.

The Auto-Poisson Probability Model

Many georeferenced datasets contain counts whose upper limit is unconstrained (e.g., lighting strikes in a given region), and hence have frequency distributions mimicking that for a Poisson random variable. Changing the probability model specification from a normal to a non-normal, one does not necessarily alter the principal impact of spatial autocorrelation on variation. The most common case of positive spatial autocorrelation also results in variance inflation for a Poisson frequency distribution, which means the chances of smaller and of extremely large values increase, at the expense of intermediate values (see Figure 2). Excessive zeroes as well as outliers may materialize. Winsorized (i.e., truncating unlikely large extreme values) spatial auto-Poisson AR models seek to furnish a description of these data when spatial autocorrelation is positive.

The common spatial Poisson regression model specification – which has Y on both sides of the equation – may be written as

$$P(Y = y_i) = c^{-1} e^{-(\mathbf{X}_i\boldsymbol{\beta} + \rho\mathbf{C}_i\mathbf{Y})} (\mathbf{X}_i\boldsymbol{\beta} + \rho\mathbf{C}_i\mathbf{Y})^{y_i} / y_i! \quad [11]$$

where y_i is the count for location i , $!$ denotes the factorial operation, $\mathbf{X}_i$ is the individual i th observation row entry in the covariate matrix $\mathbf{X}$, which includes a 1 for the constant term, and c^{-1} is the normalizing constant that functions like the Jacobian term for an auto-normal model (i.e., it ensures that all probabilities sum to 1). Unfortunately, c^{-1} is unknown, complicated, and intractable.

Fortunately, Markov chain Monte Carlo (MCMC) techniques enable the parameters of eqn [11] to be estimated. MCMC is used to simulate from a target joint auto-Poisson probability distribution, which is known only up to the normalizing constant c^{-1} . MCMC begins with the observed conditional (marginal) distributions, and outputs a sample of parameters drawn from their joint distribution. Starting with any initial conditions (e.g., independent and identically distributed Poisson random variables) and the transition matrix defined by eqn [11], the idea is to recursively and iteratively simulate each location's value using eqn [11]. Initial parameter estimates can be obtained from a conventional Poisson regression that includes the spatial lag term, $\mathbf{C}_i\mathbf{Y}$ (i.e., pseudo-likelihood estimates). An iteration involves visiting each location on a map in turn, perhaps in a random sequence, using its updated local Poisson mean to draw a

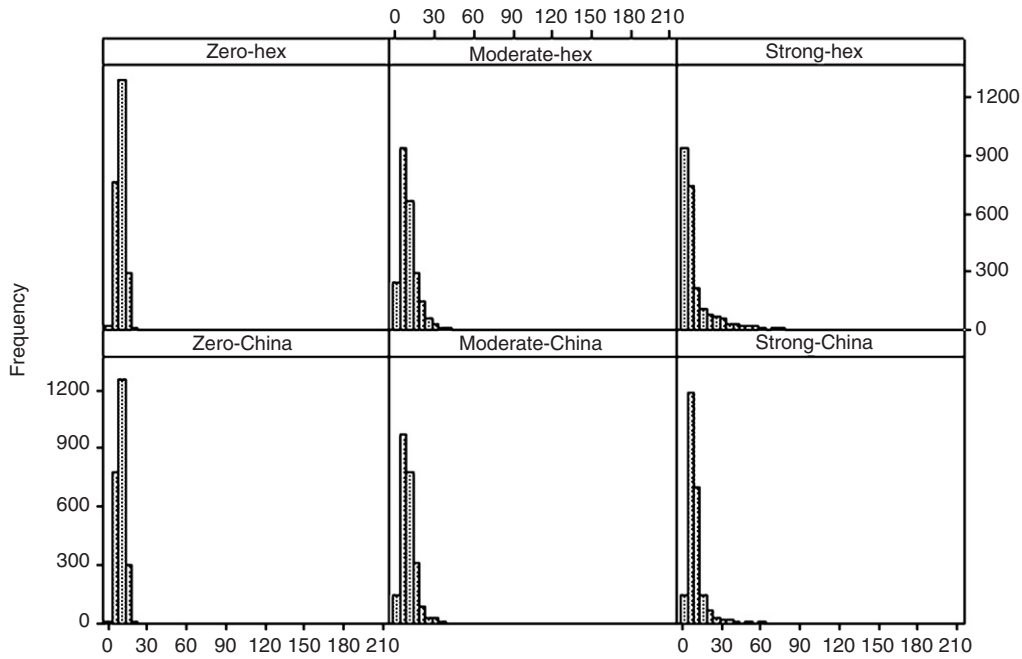


Figure 2 Histograms for simulated Poisson data containing varying levels of positive spatial autocorrelation: $\mu = 5$. Top: a 45-by-54 regular hexagonal tessellation ($n = 2430$) and bottom: China's 1990 county irregular surface partitioning ($n = 2379$).

random value from a Poisson distribution, and then immediately replacing the existing value with this new value. Winsorizing is performed when a simulated value exceeds some prespecified maximum value. Many tens of thousands of iterations must be performed. A burn-in set of beginning iterations is discarded so that the arbitrary initial map of counts becomes irrelevant. Next, time series plots of sufficient statistics need to be inspected, as do their time series correlograms, to ensure an absence of patterns and serial correlation, and convergence of parameter estimates. Frequently conspicuous patterns can be removed by increasing the burn-in period. Often serial correlation can be minimized by extracting every m th map (i.e., weeding). Convergence can be checked by generating and then evaluating several of these sequences (i.e., chains) with analysis of variance procedures. This MCMC scheme is known as Gibbs sampling, and is a method for generating a joint empirical distribution of multiple variables from a set of modeled conditional distributions for each variable when the structure of data is too complex to either implement mathematical formulas or directly simulate.

The resulting parameter estimates are obtained by constructing ratios of the converged upon distribution and the initial distribution. Constructing these ratios places the unknown normalizing constant into both the numerator and the denominator, allowing it to divide out of the problem, and hence remain unknown. Because the simple auto-Poisson model can capture only negative spatial autocorrelation, Winsorizing tends to result in $\hat{\rho}$ values

that are quite close to zero, and at best represent weak-to-moderate positive spatial autocorrelation. For corroboration purposes, these results can be approximated with an auto-normal model specification by analyzing the square root of the observed counts (the variance stabilizing Box-Cox power transformation for a Poisson variable), or the logarithmic transformation $\text{LN}(y_i + \delta)$, where δ is a translation coefficient that helps to better align these transformed values with a normal distribution.

The SF equation once again replaces $\rho\mathbf{C}$ with $\mathbf{E}_K\boldsymbol{\beta}_E$, yielding

$$P(Y = y_i) = e^{-1} e^{-(\mathbf{X}_i\boldsymbol{\beta}_X + \mathbf{E}_i\mathbf{K}\boldsymbol{\beta}_E)} \left[e^{(\mathbf{X}_i\boldsymbol{\beta}_X + \mathbf{E}_i\mathbf{K}\boldsymbol{\beta}_E)} \right]^{y_i} / (y_i!) \quad [12]$$

where the subset of eigenvectors $\mathbf{E}_K$ can be selected with a stepwise Poisson regression procedure. Some criteria suggest that this specification may well be superior to its Winsorized auto-Poisson counterpart. One appealing advantage eqn [12] exhibits over eqn [11] is that it can capture extremely high levels of positive spatial autocorrelation. Another is that it does not require MCMC techniques for parameter estimation because its normalizing constant is known and tractable; rather, it can utilize standard generalized linear modeling statistical theory and software.

The Auto-Binomial Probability Model

A number of georeferenced datasets comprise counts that have upper limits (e.g., votes by precincts for political

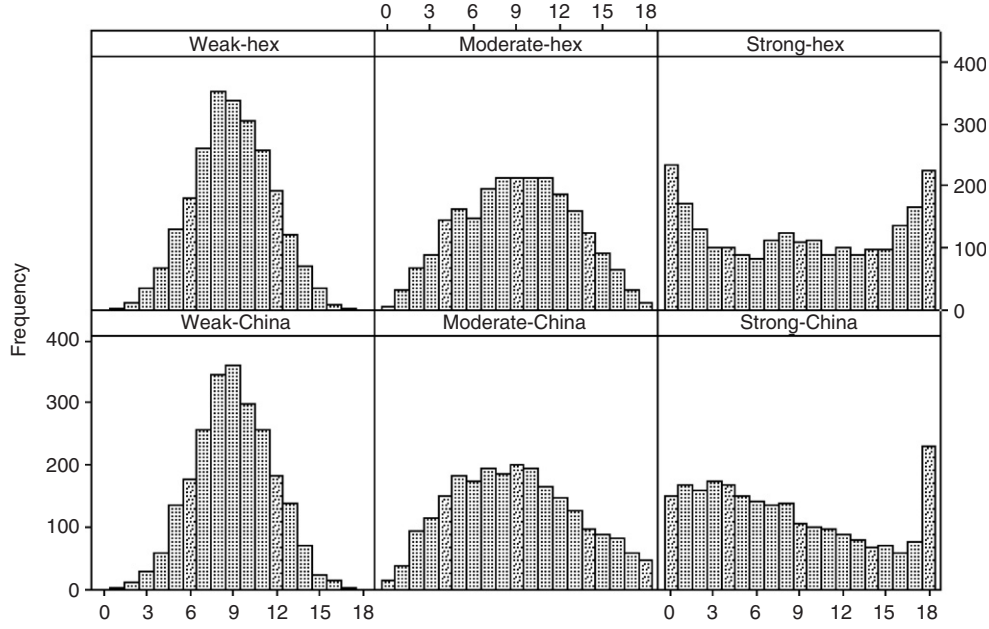


Figure 3 Histograms for simulated binomial data containing varying levels of positive spatial autocorrelation: $N=18$, $p=0.5$. Top: a 45-by-54 regular hexagonal tessellation ($n=2430$) and bottom: China's 1990 county irregular surface partitioning ($n=2379$).

candidates), and hence have a frequency distribution mirroring that of a binomial random variable. Positive spatial autocorrelation also results in variance inflation for a binomial frequency distribution, which means the chances of the extreme probabilities increase at the expense of more centrally located intermediate probabilities (see Figure 3). This effect is not detectable in binary, Bernoulli (i.e., logistic regression-type) data. Spatial auto-binomial/logistic AR models seek to furnish a description of these data.

The common spatial binomial regression model specification – which has Y on both sides of the equation – may be written as

$$P(Y = y_i) = c^{-1} \binom{N_i}{y_i} \left[\frac{e^{\mathbf{X}_i \boldsymbol{\beta} + \rho \mathbf{C}_i Y}}{1 + e^{\mathbf{X}_i \boldsymbol{\beta} + \rho \mathbf{C}_i Y}} \right]^{y_i} \times \left[1 - \frac{e^{\mathbf{X}_i \boldsymbol{\beta} + \rho \mathbf{C}_i Y}}{1 + e^{\mathbf{X}_i \boldsymbol{\beta} + \rho \mathbf{C}_i Y}} \right]^{N_i - y_i} \quad [13]$$

where N_i is the total base statistical population at location i . Again, the normalizing factor c^{-1} is unknown, complex and intractable, but as with the auto-Poisson model, parameter estimates can be obtained here with MCMC techniques.

The SF equation once again replaces $\rho \mathbf{C}$ with $\mathbf{E}_{\mathbf{K}} \boldsymbol{\beta}_{\mathbf{E}}$, yielding

$$P(Y = y_i) = \binom{N_i}{y_i} \left[\frac{e^{\mathbf{X}_i \boldsymbol{\beta}_{\mathbf{X}} + \mathbf{E}_{\mathbf{K}} \boldsymbol{\beta}_{\mathbf{E}}}}{1 + e^{\mathbf{X}_i \boldsymbol{\beta}_{\mathbf{X}} + \mathbf{E}_{\mathbf{K}} \boldsymbol{\beta}_{\mathbf{E}}}} \right]^{y_i} \times \left[1 - \frac{e^{\mathbf{X}_i \boldsymbol{\beta}_{\mathbf{X}} + \mathbf{E}_{\mathbf{K}} \boldsymbol{\beta}_{\mathbf{E}}}}{1 + e^{\mathbf{X}_i \boldsymbol{\beta}_{\mathbf{X}} + \mathbf{E}_{\mathbf{K}} \boldsymbol{\beta}_{\mathbf{E}}}} \right]^{N_i - y_i} \quad [14]$$

Now parameter estimates can be obtained with standard stepwise logistic regression or generalized linear model procedures.

Summary and Contemporary Issues

Spatial autocorrelation frequently has little impact on regression coefficient estimates for geographically distributed non-distance-based covariates, regardless of whether the statistical distribution of a variable is normal, Poisson, or binomial. In contrast, spatial autocorrelation impacts upon the variance of these variables, with positive spatial autocorrelation inducing variance inflation. This increased variance tends to flatten a bell-shaped curve, and create overdispersion in Poisson and binomial variables. Consequently, frequency distributions are distorted when spatial correlation is present. As positive spatial autocorrelation increases: (1) the center is squashed and the tails are thickened in a normal curve; (2) the shape of a Poisson distribution increasingly resembles that of an exponential distribution, with the frequency of zero and/or near-zero values increasing concomitantly with the appearance of marked outliers; and (3) the shape of a binomial distribution first approaches that of a uniform, and then that of a sinusoidal distribution. These metamorphoses are the only way variance can increase for these types of variables. In addition, spatial dependency effects compromise the descriptive ability of conventional versions of these probability models, as well as statistical inferences associated with them. Fortunately, technology now exists that enables the parameters of AR

versions of these probability models to be estimated, furnishing much better descriptions of georeferenced data.

A number of unresolved issues are raised by the foregoing discussion. Foremost is the absence of formal tests for generalized linear model residual spatial autocorrelation. Such tests also need to be developed for auto-normal residual spatial autocorrelation. Second is the lack of standard software, such as GeoBUGS for Bayesian map analysis, for implementing the necessary MCMC techniques needed to estimate the parameters of auto-Poisson and auto-binomial models. SF model specifications help circumvent this latter problem.

Nevertheless, spatial AR models have brought new power to the analysis of georeferenced data, capturing effects of redundant locational information contained in these data. Their descriptive abilities better portray observed data, and have become, and continue to be, increasingly appealing to scientists working in the spatially integrated social sciences, as well as in spatial epidemiology. They offer a more fully developed nomothetic approach that, in turn, will support an increasing emphasis on the geographically varying local statistics that fluctuate around their global parameter counterparts.

See also: Regression, Linear and Nonlinear; Spatial Autocorrelation.

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<http://www.spatial-statistics.com>
 Spatial statistics software for Matlab and Fortran by Kelly Pace, LSU.